

The music Package

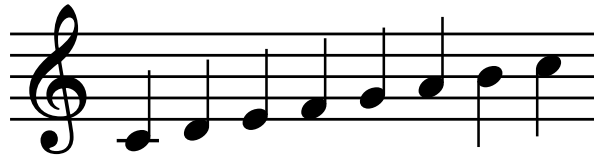
Simple staff notation, drawn from the document language

1 Reading a score

The `music` package draws a line of staff notation on a treble clef. A score is a list of notes: a letter `a–g` for the pitch, optional `'` or `,` to raise or lower it an octave, and an optional duration digit — `1` whole, `2` half, `4` quarter (the default), `8` eighth. A `|` is a barline. The heads, clefs, flags, rests, accidentals and time-signature figures are real engraved glyphs from Bravura, the reference font for music notation; the stems, beams, staff lines and ledger lines are ruled over the `\picture` layer.

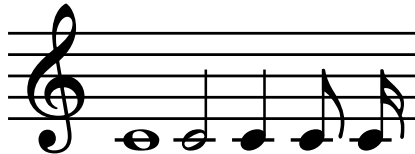
2 A scale

The C major scale, an octave of quarter notes from middle C — which sits on its own ledger line below the staff — up to the C above.



3 Durations

The same C, held for a whole note, a half, a quarter, an eighth, and a sixteenth. The longer notes have open heads; the stem appears at the half note, a flag at the eighth, and a second flag at the sixteenth.



4 Dots

A dot after the duration lengthens a note by half. It is drawn just right of the head, in the space above when the note sits on a line. Here a dotted half, a dotted quarter paired with the eighth that completes its beat, and a run of beamed sixteenths.

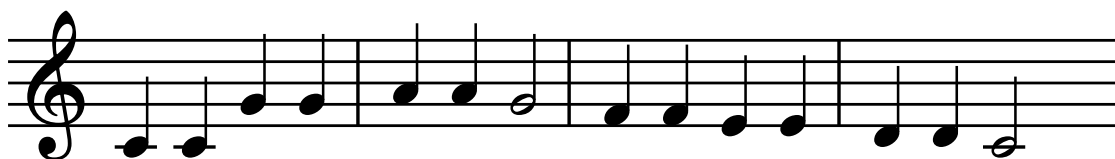


5 A melody

The opening of *Mary Had a Little Lamb*, with barlines and a half note to close each phrase.

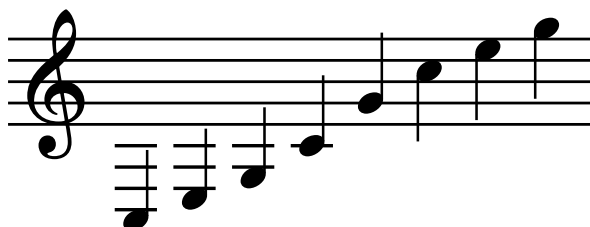


And the start of *Twinkle, Twinkle, Little Star*.



6 Range and ledger lines

Notes far above or below the staff hang on short ledger lines, drawn only as far as the note reaches.



7 Accidentals

An accidental leads a note: + for a sharp, - for a flat, n for a natural. The sign is drawn just left of the head; it changes how the note is read, not where it sits, so +f and f share a line. Here a chromatic climb, then a sharpened note cancelled by a natural.

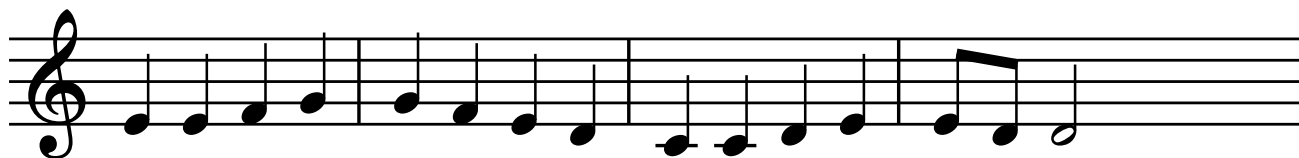


8 Beams

Eighth notes in a run are usually joined by a beam rather than each carrying its own flag. Bracket a group with [and] and the notes between share one beam.



A beamed phrase mixed with a longer note — the opening of *Ode to Joy*.



9 Time signatures

Setting `\set musictimenum` and `\set music timeden` draws a time signature after the clef. Three-four time, a waltz, with the beats grouped into bars.



10 Key signatures

A key signature follows the clef. Set `\set musickey` to a signed count: a positive number draws that many sharps, a negative one that many flats, each in its conventional order and on its conventional line. Here two sharps (D major), then three flats (E-flat major).



11 Repeats and barlines

Besides the plain barline |, a score can write a double barline ||, a final barline |., and the start and end of a repeated section, |: and :|.



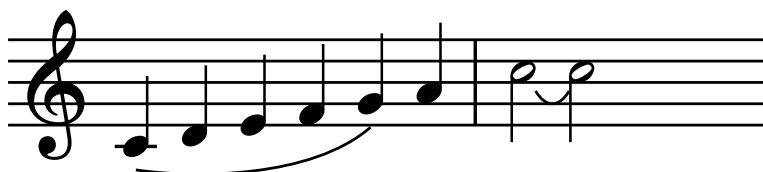
12 Articulations

A mark after a note tells the player how to attack it: > an accent, ! a staccato, = a tenuto, * a marcato, ; a fermata. Each sits on the side away from the stem, and several may stack. Here the same phrase plain, then accented, staccato, tenuto and marcato, closing on a fermata.



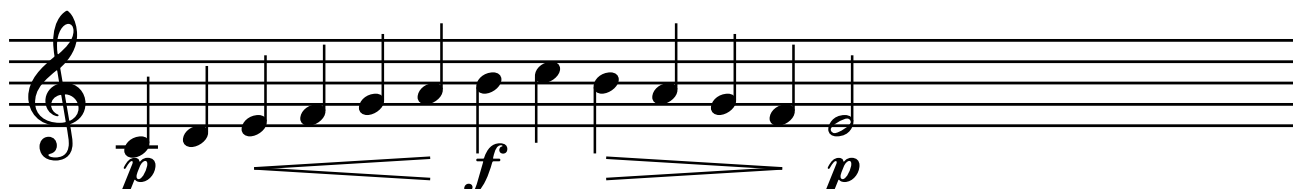
13 Slurs and ties

A run bracketed by (and) is drawn under one slur. A trailing - ties a note to the next of the same pitch, holding it across. Here a slurred turn, then two tied half notes.



14 Dynamics and hairpins

A token beginning ! is a dynamic, set below the staff: !p, !mf, !ff, !sfz. A < opens a crescendo hairpin and > a diminuendo, each closed by =. Here a phrase that starts soft, swells, and falls away.



15 The bass clef

Setting `\set musicclef {bass}` notates in the bass clef instead, where middle C sits one ledger line *above* the staff. Here a C major scale from the C below the staff up to middle C.



16 Alto and tenor clefs

The C clef names the line that is middle C: the middle line in the `alto` clef, the fourth line in the `tenor`. The same C major scale around middle C, read in each.



17 Choosing the music font

The glyphs come from a SMuFL music font, and `\set musicfont` chooses which one. The default is Bravura, the standard reference font. Setting it to `petaluma` switches to a handwritten, jazz-lead-sheet style — the same notes, a different hand. Here the C major scale again, in Petaluma.

